

Question	Answer	Marks
5(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out junction or (algebraic) sum of current(s) at a junction is zero	B1
5(b)(i)	$(V =) E - Ir$	C1
	$(V =) 5.0 - (38 \times 10^{-3} \times 4.7) = 4.8 \text{ (V)}$	A1
5(b)(ii)	$R_{(Z)} = 120 \, \Omega$	C1
	$I_2 = 38 \times 10^{-3} - (4.8 / (100 + 120))$	C1
	$I_2 = 0.016 \text{ A}$	A1
	or	
	$R_{(Z)} = 120 \, \Omega$	(C1)
	$R_{(\text{EXT})} = 4.8 / 38 \times 10^{-3}$ $(= 126 \, \Omega)$ $1 / R_{(Y)} = 1 / 126 - 1 / (120 + 100)$ $(R_{(Y)} = 297 \, \Omega)$ $I_2 = 4.8 / 297$	(C1)
	$I_2 = 0.016 \text{ A}$	(A1)

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5(b)(iii)	$P = IV$ or $P = I^2 R$ or $P = V^2 / R$	C1
	$P = 0.016 \times 4.8$ or $P = 0.016^2 \times (4.8 / 0.016)$ or $P = 4.8^2 / (4.8 / 0.016)$ $P = 0.077 \text{ W}$	A1
5(b)(iv)	resistance of the LDR / Z increases (as light intensity decreases)	B1
	<u>total</u> resistance (of the circuit) increases or current <u>in the battery</u> decreases	M1
	(potential difference across the internal resistor decreases so) the terminal potential difference increases	A1

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